GRADUATE CERTIFICATE IN PATTERN RECOGNITION SYSTEMS

AValidate

Intelligent Audio-Visual Source Matching

Does the thing you see make the sound you hear?

FIRST PROJECT PRESENTATION

Group 25 · 6 Oct 2026

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Problem & Motivation

In real-world video, the object on screen is sometimes not the source of the sound.

SEEN

Dog

HEARD

Barking

Strong match

The visible dog is the source.

SEEN

Car

HEARD

Car horn

Strong match

The visible car is the source.

SEEN

Dog

HEARD

Siren

Mismatch

The siren comes from somewhere off-screen.

SEEN

Person

HEARD

Dog bark

Uncertain

A dog may be off-screen, or the person may be imitating one.

RESEARCH QUESTION

Can synchronised audio and visual embeddings be fused to reliably verify whether the visible event is the source of the observed sound?

Why It Matters

When a system pins a sound on the wrong thing, real mistakes follow. Four examples.

Video search & tagging

THE PROBLEM

A video gets tagged “dog” because a dog barks, but no dog ever appears. People searching for dog videos get the wrong clip.

WHAT AVALIDATE DOES

Checks that the dog is actually on screen before the tag is kept.

Training data for AI

THE PROBLEM

In many online clips the sound comes from off-screen. Models trained on them learn the wrong links.

WHAT AVALIDATE DOES

Flags those clips automatically, so they can be removed before training.

Edited & re-dubbed video

THE PROBLEM

During editing, audio gets swapped or slips out of sync. The sound no longer matches the picture.

WHAT AVALIDATE DOES

Shows the exact seconds where sound and picture stop matching.

Security cameras

THE PROBLEM

A camera picks up glass breaking and the system blames whoever is in view, even though the sound came from the next room.

WHAT AVALIDATE DOES

Says “uncertain” when the person on screen is probably not the source.

Same question every time: is the thing on screen actually making the sound?

Project Objectives

Five things the system must do, in the order the data flows through it.

01

Extract & align

Pull synchronised audio and video frames from each clip and align them at about one-second resolution.

02

Represent

Encode each modality into embeddings, starting from lightweight baselines and moving to pretrained encoders.

03

Recognise

Predict the event class, and whether the event is both audible and visible in each second.

04

Verify

Learn a correspondence score that decides match or mismatch between what is seen and what is heard.

05

Present

For an uploaded clip, show the event, the decision with its confidence, and the per-second evidence in a simple demo.

How we judge success: whether the fused model separates matched from mismatched clips better than two independent classifiers whose labels are simply compared, including on hard, similar-sounding mismatches.

Project Scope

An offline MVP first. Extensions only once it is stable.

MVP · COMMITTED

- Offline 10-second clips from the AVE dataset
- 28 event categories with per-second labels
- Audio branch, visual branch, fusion and three prediction heads
- Match or mismatch, reported as Strong, Weak/Uncertain or Mismatch
- A demo that accepts a clip and shows the result with a timeline

EXTENSIONS · IF TIME ALLOWS

- Attention-based fusion, compared with the simple fusion
- Scale-up with a VGGSound subset of overlapping classes
- Qualitative sound-source localisation on VGG-SS
- Robustness tests: volume, background noise, partial occlusion
- Live webcam and microphone mode for the final demo

BOUNDARIES

- A low score means weak evidence of correspondence, never proof that the pairing is impossible
- Events outside the 28 trained categories are not recognised
- Off-screen sources are flagged as uncertain rather than located
- Localisation, if shown, is illustrative and not scored

Data Required & Available

One primary dataset, one generated set, two optional sources. All public.

PRIMARY

AVE · Audio-Visual Event

- 4,143 ten-second YouTube clips
- 28 event categories
- Per-second event / background labels
- Official split: 3,339 train · 402 val · 402 test

BUILT BY US

Generated pairs

- Positives: original synchronised clips
- Negatives: audio swapped in from other clips
- Easy, hard and time-shifted negatives
- Generated inside each split

MIGHT USE

VGGSound

- Large-scale audio-visual clips
- Subset of classes that overlap with AVE
- Scale-up, or fallback if AVE is too small

MIGHT USE

VGG-SS

- Sound-source localisation annotations
- Qualitative checks of where the sound comes from
- Not used for training

PREPROCESSING PIPELINE

1. **Extract audio**
   mono, one fixed sample rate
2. **Sample frames**
   at timestamps aligned to each second
3. **Map labels**
   per-second event / background
4. **Build pairs**
   positives and three kinds of negative
5. **Store splits**
   reproducible, versioned train / val / test

No leakage: every clip belongs to one split, and negatives only borrow audio from clips in the same split.

Training Pairs

The model learns correspondence from real clips and from deliberately broken ones.

POSITIVE

VIDEO



Dog video

AUDIO



Its own barking

Label **MATCH**

The original, synchronised clip.

EASY NEGATIVE

VIDEO



Dog video

AUDIO



Siren from another clip

Label **MISMATCH**

Different category. Easy to reject.

HARD NEGATIVE

VIDEO



Guitar video

AUDIO



Banjo from another clip

Label **MISMATCH**

Similar-sounding category: guitar, banjo, ukulele, mandolin; truck and bus.

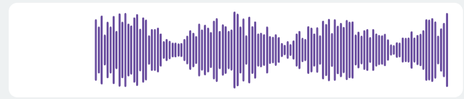
TIME-SHIFTED

VIDEO



Dog video

AUDIO



Its own audio, 3 s late

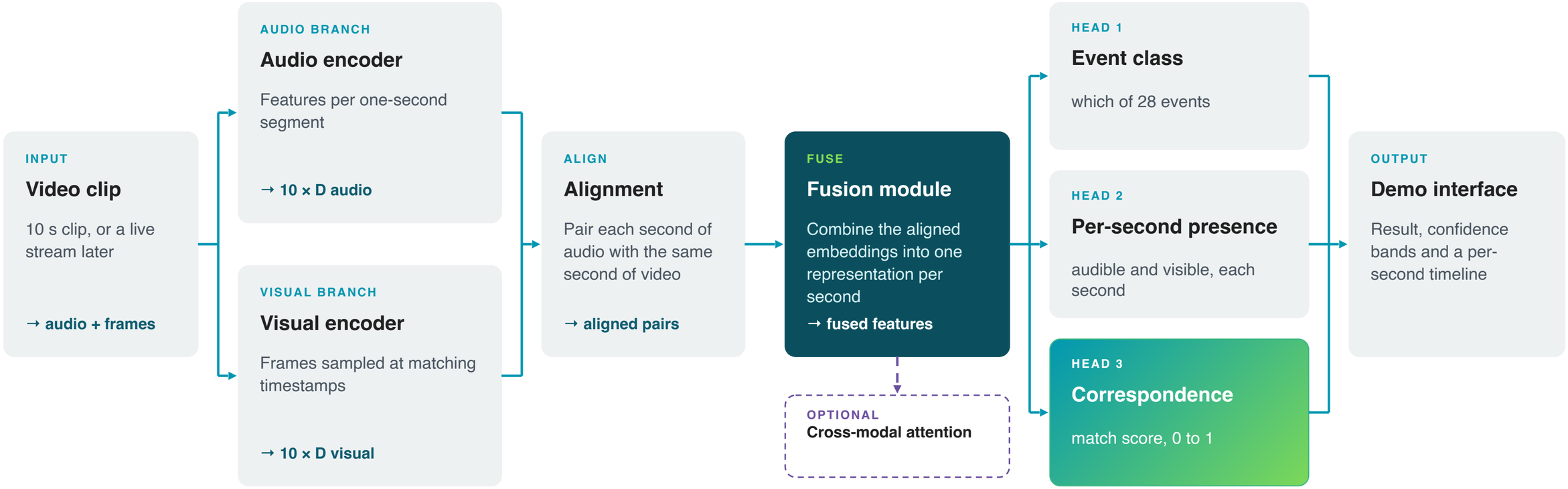
Label **MISMATCH**

Right sound, wrong moment. Tests timing.

Why the hard cases matter: with easy negatives alone, a model can pass by checking whether two category labels agree. Similar sounds and time shifts make it use the actual audio-visual evidence. Audio swapping follows Look, Listen and Learn [6]; time shifts follow [7][8].

System Design

Two branches encode each stream on its own; fusion decides whether they belong together.



Shared interface: both branches emit one embedding per second on the same timestamps, keyed by clip ID. Each branch can be swapped or improved without touching the rest.

ML Techniques

Each part starts with a simple baseline, so every improvement can be measured against it.

Audio branch

BASELINE

Log-Mel spectrogram with a small CNN

CANDIDATE

Pretrained audio encoder, e.g. BEATs [4], adapted with LoRA

OUTPUT

One audio embedding and event evidence per second

Visual branch

BASELINE

Frame features from a pretrained image CNN

CANDIDATE

Pretrained vision-language encoder, e.g. CLIP ViT-B/16 [5], adapted with LoRA

OUTPUT

One visual embedding per second

Fusion

FIRST VERSION

Concatenate aligned embeddings, then an MLP (late fusion)

UPGRADE

Cross-modal attention, where each stream attends to the other

OUTPUT

One fused representation per second

Prediction heads

EVENT

Softmax over 28 categories

PRESENCE

Per-second audible-and-visible classifier

CORRESPONDENCE

Match score, calibrated, then banded into Strong, Weak/ Uncertain or Mismatch using thresholds chosen on validation data

Named models are indicative. Final choices depend on what the experiments show.

Evaluation Plan

Five experiments, fixed before any model is trained. Each answers one question.

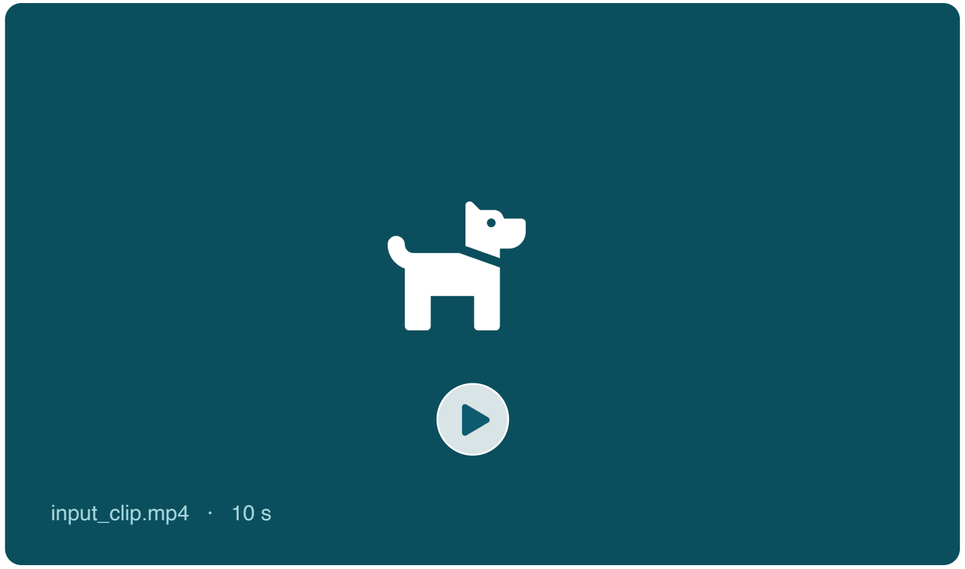
#	EXPERIMENT	QUESTION IT ANSWERS	COMPARISON	MEASURES
1	**Unimodal event recognition**	Does each branch recognise the event on its own?	Lightweight baseline vs. pretrained / adapted encoders	**Accuracy, Macro-F1, confusion matrix**
2	**Audio-visual correspondence**	Does fusion add anything over two separate classifiers?	Label-agreement baseline vs. learned fusion (simple vs. attention)	**ROC-AUC, PR-AUC, F1 at a validation-selected threshold**
3	**Negative difficulty**	Does it still work when the mismatch is subtle?	Easy vs. hard (similar-category, time-shifted) negatives	**ROC-AUC per negative type**
4	**Temporal presence**	Does it find the right seconds?	Per-second predictions vs. AVE event / background labels	**Per-second F1**
5	**Confidence reliability**	Can the confidence be trusted?	Raw vs. calibrated correspondence scores	**Expected Calibration Error, reliability diagram**

Scenario tests dog + bark · car + horn · dog + siren · person + dog bark **Robustness** score drop as volume, noise and occlusion vary

Headline result: correspondence ROC-AUC on hard negatives, compared with the label-agreement baseline.

What the User Sees

One clip in. The event, a decision with confidence, and the seconds that support it.



DECISION

STRONG MATCH

0.91 correspondence

Event Dog bark (0.88)

CONFIDENCE BANDS

Strong match
score ≥ upper threshold

Weak / Uncertain
between the thresholds

Mismatch
score < lower threshold

Thresholds are chosen on validation data.

PER-SECOND EVIDENCE



Illustrative mock-up. No results yet.

Progress So Far

Design and planning are complete. Build starts with data and baselines.

DONE

- ✓ Problem and research question defined
- ✓ AVE selected as the primary dataset; optional sources identified
- ✓ System design and prediction heads agreed.
- ✓ Evaluation plan fixed, including the label-agreement baseline and hard negatives
- ✓ Shared GitHub repository set up; work split by modality (Person Assignment is left), each member owning data and model

NEXT

1. **Data**
   Download AVE, extract audio and frames, map per-second labels
2. **Baselines**
   Spectrogram CNN for audio; image-CNN features for video
3. **Pairs & harness**
   Pair generator and a shared evaluation script
4. **Fusion v1**
   Late fusion with the three heads
5. **Demo**
   First end-to-end run on a single clip

Team & Roles

Each member owns one slice from raw data to model output. Integration is shared.

01

Member 1

Audio

DATA

Audio extraction and preprocessing; per-second audio labels

MODEL

Audio baseline and pretrained encoder; audio experiments

02

Member 2

Vision

DATA

Frame extraction and sampling at aligned timestamps

MODEL

Visual baseline and pretrained encoder; robustness tests; demo

03

Member 3

Fusion & verification

DATA

Pair generation; evaluation and calibration data

MODEL

Fusion, correspondence scoring, calibration; final integration

Shared by all three: integration, end-to-end evaluation, and one inference function. A video goes in; the event, confidence values, per-second evidence and match score come out.

Thank you

Questions and feedback welcome

AValidate · Group 25

Appendix References & AI Use

REFERENCES

[1] Tian, Y. et al., “Audio-Visual Event Localization in Unconstrained Videos,” ECCV, 2018.

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[5] Radford, A. et al., “Learning Transferable Visual Models From Natural Language Supervision,” ICML, 2021.

[6] Arandjelović, R. and Zisserman, A., “Look, Listen and Learn,” ICCV, 2017.

[7] Owens, A. and Efros, A. A., “Audio-Visual Scene Analysis with Self-Supervised Multisensory Features,” ECCV, 2018.

[8] Korbar, B., Tran, D. and Torresani, L., “Cooperative Learning of Audio and Video Models from Self-Supervised Synchronization,” NeurIPS, 2018.

USE OF AI TOOLS

ChatGPT and Claude were used to help review and refine the proposed architecture, structure the proposal and these slides, improve technical expression, and prepare diagrams. All model choices, dataset selections, implementation decisions, experiments and final content are reviewed and approved by the project team.